An Offline Speech Recognition-Assisted Application for Chinese and Other Language Learning

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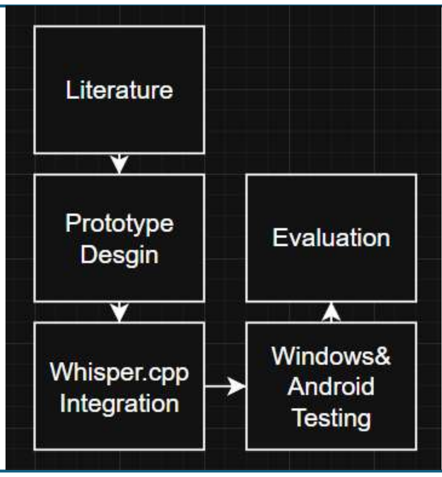
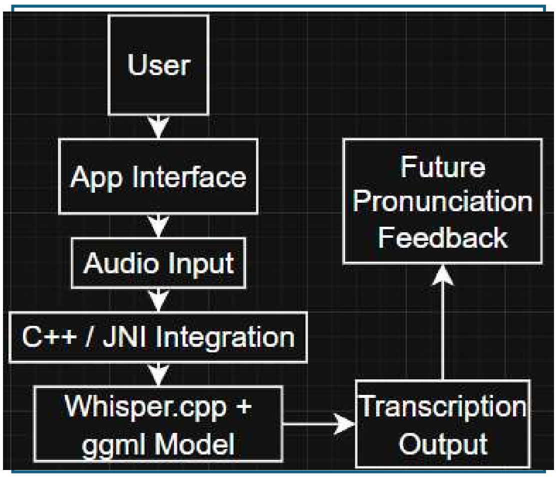
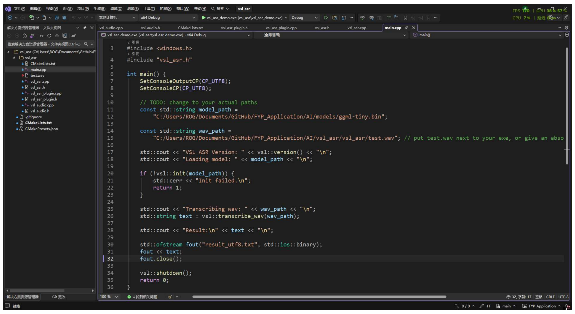
Chinese pronunciation is difficult for beginners because Mandarin is a tonal language. Many existing learning apps provides simple "correct" or "incorrect" feedback most. This project explores offline speech recognition as a foundation for future pronunciation feedback.

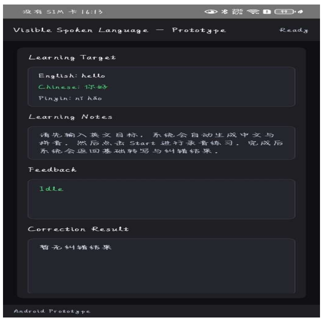
- **Existing apps provide limited pronunciation feedback.**
- **Cloud-based recognition may cause privacy, latency, and internet-dependency issues.**
- **Beginners need clearer and faster support when practising pronunciation independently**

Aim: To build an offline speech recognition prototype for Chinese language learning and future pronunciation feedback.
Object: Use whisper.cpp for local speech recognition.
Create a basic recording and transcription interface.
Test phrase-level Chinese recognition.
Explore Android deployment.
Identify limitations and future improvements.

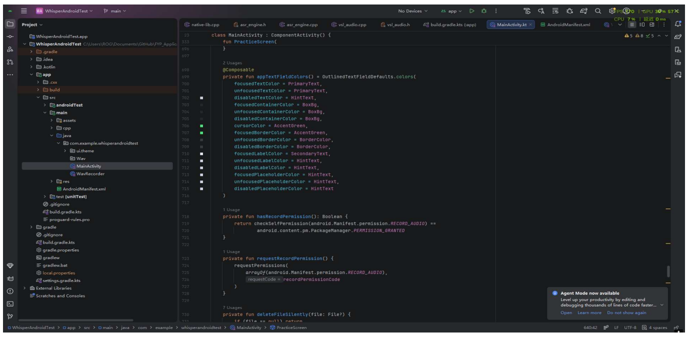








- **The prototype can load local speech recognition models.**
- **Basic Chinese phrase-level recognition was achieved.**
- **A Windows prototype and a basic Android version were implemented.**
- **Offline recognition was feasible, but performance depends on device environment and noise level.**



The project shows that offline ASR can support privacy-friendly and low-latency language learning. However, the current version is still a technical prototype, not a complete pronunciation-learning product. The main challenges were Android integration, model loading, dependency linking, and noise sensitivity.

This project successfully developed a working offline speech recognition prototype for Chinese language learning. Will add pronunciation scoring and tone feedback. Also, will improve Android performance.

Radford et al. (2023) / ggml-org (2026) / Han et al. (2019) /Farran & Morett (2024)

Acknowledgements | GitHub | QR Code